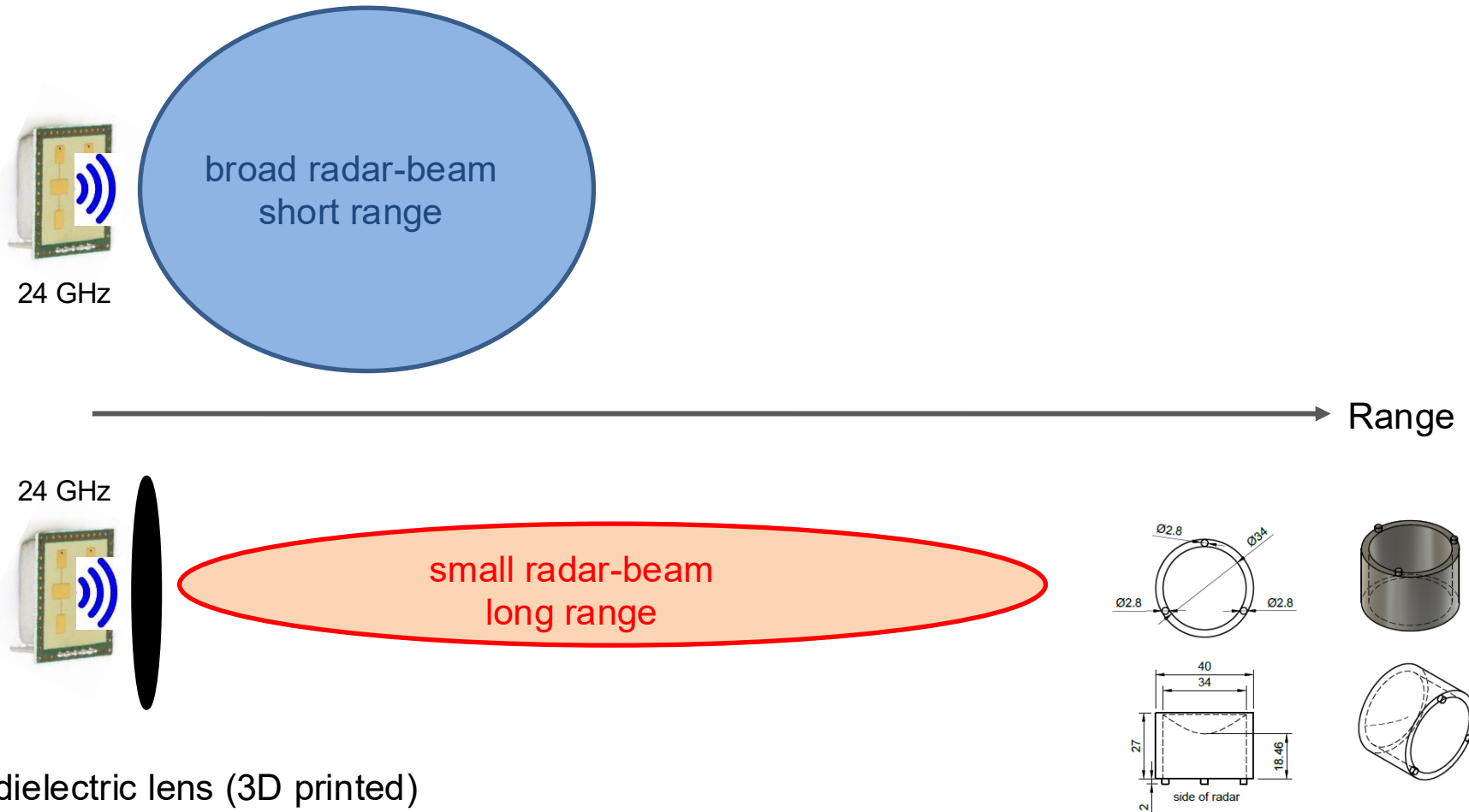


Option Dielectric Lens Antenna



- design with ordinary formulas from optics
- R.J. Friel et al., "3D Printed Radar Lenses with Anti-Reflective Structures", MDPI, 2019.

Speed of Light and Refractive Index

- Speed of electro-magnetic waves (**speed of light**) in **vacuum**

$$c = \frac{1}{\sqrt{\epsilon_0 \cdot \mu_0}}$$

- Speed of electro-magnetic waves in a **medium**

$$v = \frac{1}{\sqrt{\epsilon_0 \cdot \epsilon_r \cdot \mu_0 \cdot \mu_r}}$$

In a medium ($\epsilon_r > 1$ and $\mu_r \geq 1$) the speed is lower

- Definition of Refractive Index

$$n = \frac{c}{v} = \frac{1}{\sqrt{\epsilon_r \cdot \mu_r}}$$

This is well known and applied in optics

- Snell's law $n_1 \cdot \sin\theta_1 = n_2 \cdot \sin\theta_2$ is used for optical systems
- Radar signals are electro-magnetic waves

Lenses of material with $\epsilon_r > 1$ can be used for focusing

